Project Objectives & Tech Stack

1. Project Objectives

The primary objective of the Store Sales Forecasting project is to build a robust and scalable machine learning pipeline capable of accurately predicting future sales for various store items across different families and store numbers.

Key Objectives:

1. Demand Forecasting: Predict future sales over a specific horizon based on historical data, enabling better inventory management.

2. Pattern Recognition: Identify seasonality, trends, and the impact of promotions and holidays on product sales.

3. Automated Pipeline: Implement an end-to-end ML pipeline encompassing data ingestion, preprocessing, training, and prediction.

4. Scalable Deployment: Serve the machine learning models via a scalable REST API (FastAPI) to allow seamless integration with frontend or external applications.

5. Reproducibility: Ensure that all experiments, datasets, and models are tracked and versioned using Data Version Control (DVC).

2. Tech Stack Used

The project leverages a modern Python-based Data Science and MLOps tech stack.

Machine Learning & Data Processing:

- Python 3: Core programming language.

- Pandas & NumPy: For data manipulation, cleaning, and feature engineering.

- LightGBM & Scikit-learn: High-performance gradient boosting frameworks for training the predictive

models.

- Darts: A specialized Python library for user-friendly forecasting on time series.

- Hampel: Used for outlier detection and removal in time-series data.

API & Deployment:

- FastAPI: A modern, fast web framework for building APIs with Python.

- Uvicorn: An ASGI web server implementation for Python.

- Docker: For containerizing the application, ensuring consistency across environments.

MLOps & Data Versioning:

- DVC (Data Version Control): To version datasets and track ML models.

- Joblib: For saving and loading the trained LightGBM models efficiently.

Database:

- PyMongo: MongoDB driver for Python, potentially used to store raw sales records, predictions, or application logs.

3. Conclusion

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This stack is perfectly suited for scalable time-series forecasting. It provides the necessary tools to handle complex patterns in sales data (like promotions and holidays) and offers a reliable way to serve predictions in real-time.